2 A business sells a single product. Customers can purchase one or more of this product.

Each sale has an ID and a quantity, for example "ABC" and 2

The business needs a program to store the data about the sales in a circular queue data structure.

(a) Write program code to declare a record structure, `SaleData`, to store the data about each sale.

Save your program as **Question2_J2023**.

Copy and paste the program code into **part 2(a)** in the evidence document.

[2]

(b) Write program code to:

- declare a global array, `CircularQueue`, of 5 items to store the sale records
- declare the global pointers `Head` and `Tail`
- declare the global variable `NumberOfItems`
- initialise all elements of the array `CircularQueue` to an empty record, where the ID is null (`""`) and quantity is `-1`
- initialise `Head`, `Tail` and `NumberOfItems` to `0`

Save your program.

Copy and paste the program code into **part 2(b)** in the evidence document.

[4]

(c) The function `Enqueue()`:

- takes a new record as a parameter
- inserts the record in the circular queue at the element pointed to by `Tail`
- updates pointers and other variables as required
- returns `-1` if the circular queue is full
- returns `1` if the record is stored successfully.

Write program code for the function `Enqueue()`.

Save your program.

Copy and paste the program code into **part 2(c)** in the evidence document.

[6]

(d) The function `Dequeue()`:

- returns a null or empty record if the circular queue is empty
- returns the first record in the queue if the circular queue is not empty
- updates pointers and other variables as required.

Write program code for the function `Dequeue()`.

Save your program.

Copy and paste the program code into **part 2(d)** in the evidence document.

[6]

(e) The procedure `EnterRecord()`:

- takes an ID and quantity as input and creates a sale record
- uses `Enqueue()` to insert the record in the circular queue
- outputs `"Full"` if the record was not inserted in the circular queue
- outputs `"Stored"` if the record was inserted in the circular queue.

Write program code for the procedure `EnterRecord()`.

Save your program.

Copy and paste the program code into **part 2(e)** in the evidence document.

[5]

(f) The following sale records need to be entered into the program:

ID	Quantity
ADF	10
OOP	1
BXW	5
XXZ	22
HQR	6
LLP	3

(i) Amend the main program to:

- use `EnterRecord()` to input the six records in the table
- use `Dequeue()` to remove one record
- output either the ID and quantity of the removed record, or an error message if the circular queue is empty
- use `EnterRecord()` to input the record with the ID "LLP" for a second time
- output the ID and quantity for all the records currently stored in the array `CircularQueue`.

Write program code to perform these tasks.

Save your program. Copy and paste the program code into **part 2(f)(i)** in the evidence document.

[4]

(ii) Test your program.

Take a screenshot of the output.

Save your program. Copy and paste the screenshot into **part 2(f)(ii)** in the evidence document.

[1]